Atlas 210X — VFO Nano add-on wiring

YT3CT (ex YU5KBM) · Nano ATmega328P (Old Bootloader) · sketch atlas_vfo_fixed.ino · 2026-10 · button / relay part = proposal

1. LCD RG1602A (16×2) — twisted wiring, 12 wires

Twisted group	Nano board pad	LCD pin
orange / black	orange: D8 → 470 Ω	6 E
	black: GND	1
blue / black	blue: D7 → 470 Ω	4 RS
	black: GND	1
red / black / yellow / grey	red: +5 V	2 VDD
	black: GND	1
	yellow: contrast pot wiper	3 V0
	grey: 10 Ω pad (backlight)	15 A
brown / white	brown: D9 → 470 Ω	11 D4
	white: D10 → 470 Ω	12 D5
purple / green	green: D11 → 470 Ω	13 D6
	purple: D12 → 470 Ω	14 D7

- **Nano board:** pads unchanged; all 3 black wires on the one GND pad. 470 Ω in line within ~2 cm of the pads (6 signal wires only).
- **At the LCD:** existing jumper 16–5–1 stays; 100 nF + 10 µF across pins 2–1. Pins 7–10 open.
- Different twist per group, < 1 cm untwisted at the ends. Groups side by side along the chassis.

2. TTL programming leads (no USB needed)

USB-TTL cable (Prolific)	Nano
Green (TX)	D0 / RX
White (RX)	D1 / TX
Black (GND)	GND
Red (+5 V)	**NOT connected**
—	RST lead (+ optional push button to GND)

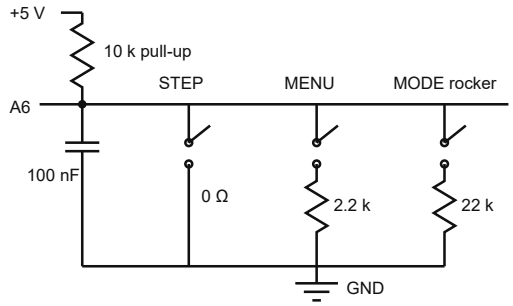
- No sync: swap green / white. **Upload:** start it, then RST to GND for ½ s (VariBeam Nano the same).
- Power cycle only with the POW LED fully out (TTL back-feeds the Nano). Never USB and TTL at the same time.

Free Nano pins & plan proposal

Pin	Now	Plan
D0 / D1	Serial	TTL / J4 serial
D4	free	AGC fast switch sense
D5	free	ATT relay sense
D6	STEP button	PRE relay sense (STEP → A6)
D13	PTT request	→ 10 k → VariBeam Nano D8 (300 ms filter: bootloader blinks ignored)
A6	free (analog)	STEP, MENU + MODE rocker
A7	free (analog)	S3 position: CW / CAL sense

Used: D2/D3 encoder, D7–D12 LCD, A0/A1 S-meter, A3 band, A4/A5 I^2C Si5351. **A2 TX** (LOW = TX): KEY line → 1N4148 (band to key line) → A2. **Sense inputs:** GND-side switch + 1N4148, or 22 k / 10 k from +12 V. Parts at the **source end** (wires to the Nano carry ≤ 5 V). Never raw +13 V on a pin.

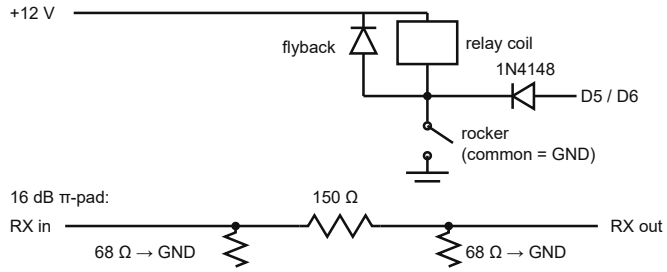
3. STEP / MENU buttons + MODE rocker on A6 proposal



Control	R to GND	ADC	Accept if
STEP button	0 Ω (direct)	0	< 120
MENU button	2.2 k	184 (170 with MODE)	120–480
MODE rocker (OPP)	22 k	703	480–860
nothing	—	1023	> 860

A6/A7 have **no internal pull-up** → the external 10 k is required. MODE is latching (NORM / OPP like the original S1): its state is taken while no button is pressed. 5 % resistors fine.

4. ATT / NORM / PRE — ON-OFF-ON rocker, DPDT relays, RX only proposal

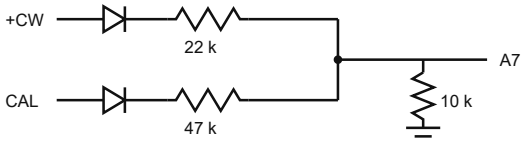


- Coil high end on +12 V, the rocker switches the coil low end to GND (ATT or PRE, centre = NORM). The 1N4148 (band to the coil) passes the LOW to the Nano pin and blocks the +12 V when off.
- DPDT signal relays: both poles switch the RF (in and out). π-pad 68 / 150 / 68 Ω ≈ 16 dB (20 dB: 62 / 240 / 62 Ω). Full board: preamp_bfg135.pdf.
- **Only in the receive line after the antenna relay — never in the TX path.** Short leads / thin coax.

5. AGC fast/slow — DPDT rocker, sense D4 proposal

- 2N7000 at PC-300D: gate via 100 Ω from the rocker (+12 V), 100 k gate–GND, source GND, drain via R_{fast} (~150–330 k, to try) to the AGC hold node on PC-300D.
- Parts on PC-300D to be identified by value first (hold cap / 1 M discharge R).
- Sense: **DPDT rocker** — 2nd pole common to GND, "fast" throw to D4 (INPUT_PULLUP, LOW = fast). No divider.

6. S3 CW / CAL sense on A7 proposal



S3	A7 (13 V)	Nano
CW	≈ 3.8 V	BFO ~800 Hz into the passband, LO follows
CAL	≈ 2.1 V	CLK2 calibrator on
REC / TRANS	0 V	normal

MODE rocker NORM / OPP = sideband as the Atlas S1: software only (BFO to the other side of the crystal filter, 5.520.0 ↔ ~5.523.3 MHz, LO follows).

7. Front panel gain controls — 2 concentric dual pots, fixes fitted 2026-10

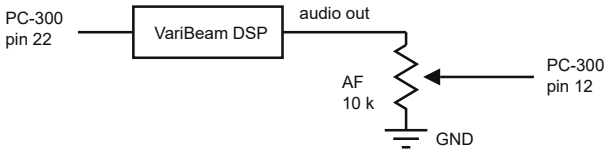
Pot	Function	Orig.	Fitted	Fix
1	AF gain (R4)	10 k	10 k	none
	RF gain (R6)	10 k	5 k	R5 4.7 k → 2.2 k, R7 470 → 220 Ω
2	MIC gain (R12)	10 k	20 k	20 k across outer lugs
	ALC (R11)	10 k	100 k	11 k across outer lugs

- RF gain: GND–R5–pot–R7–+8.5 V, wiper to PC-100 pin 14; wiper range ~2.6–8.2 V again. Measure R5 first (if already ~2.4 k, only R7).
- Resistor across the outer lugs = 10 k load as original, wiper ratio unchanged. 100 k ALC = harder, slower ALC → first suspect for low modulation.

8. J4 rear socket (old EXT OSC) — CAT / PTT / audio proposal

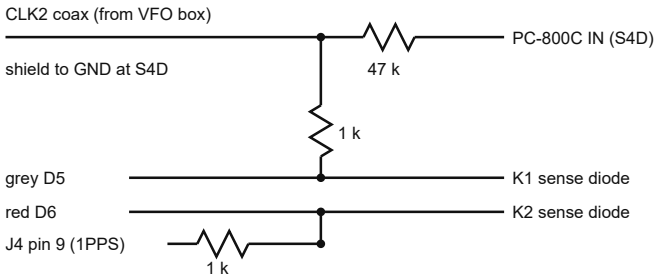
Pin	Use
1	RST (grey) — **1 µF (105) in series** between pin 1 and Nano RST, at the Nano end (not to GND). 10 nF is too small: the Nano's own 100 nF on RST swallows the pulse. From adapter DTR (programming from the rear)
2	Nano TX D1 (green) → adapter RX
3	Nano RX D0 (white) ← adapter TX
4	GND (black, chassis / Nano GND)
5	Audio in (PC → TX): transformer → 47 k / 470 Ω → PTT relay **NO** → COM → PC-200 pin 17 (MIC gain controls it)
6	Audio GND, PC side — **floating**, open to chassis
7	Audio out (RX → PC): PC-300 pin 22 → 1 µF → 10 k → transformer (AF gain has no effect)
8	spare (PTT is now CAT → Nano, section 10)
9	GPSDO 1PPS (AUTO CAL, planned) → 1 k → red D6 wire; plug in only to calibrate

- Old wires (2–3 jumper, 5 +8.5 V, 6 stub, 7 T line, 8 +13 V fused) removed, GND on 4 kept. Check REC / SSB TX / CW after.
- USB-TTL adapter outside. Audio via 600:600 transformers, shielded, away from the serial pair. RX audio from PC-300 pin 22 (before the VariBeam).
- PTT = **CAT command** (e.g. Kenwood TX; / RX;) over the same TTL adapter → VFO Nano D13 HIGH → VariBeam Nano D8; D8 HIGH ≥ 300 ms → D9 → BC547 → PTT relay (section 10). One USB cable for CAT, PTT and uploads. Debug serial off with CAT; WSJT-X / MSHV Tx delay ≈ 0.5 s.



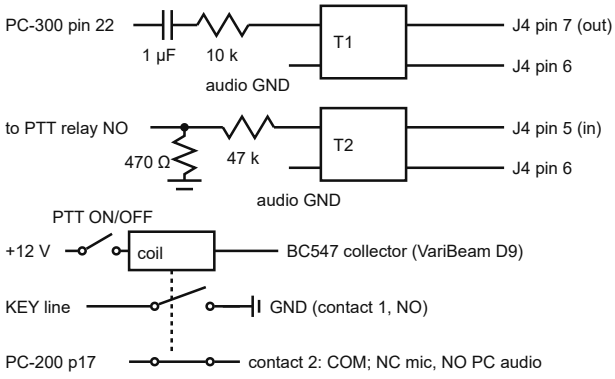
- AGC / S-meter pickoff is before pin 22. Shielded audio leads, shield grounded at one end.
- R13 10 k (mic gain top) goes to the WHT/BLK **+CW** line (S3 CW / PC-100 pin 18), 0 V in SSB. Not PC-100 pin 21 (+13 V on every TX).

9. Calibrator + AUTO CAL — Si5351 CLK2 CLK2 wired



- **Wired 2026-10-26:** CLK2 coax centre → 47 k → PC-800C IN (S4D, where the PC-820 output went = RX side of the antenna relay, cut off on TX), shield to GND there. Marker passes the whole RX chain incl. ATT/PRE. PC-820 removed, its CAL supply wire insulated.
- CLK2 on only in S3 CAL (A7 sense) or from the menu — off otherwise, disabled output **high-Z**. Marker level: CLK2 drive / larger series R if too strong.
- **AUTO CAL (planned, not wired yet):** 1 k from the coax centre at S4D to a tap on the grey D5 wire (Timer1 counter input); J4 pin 9 → 1 k → tap on the red D6 wire (1PPS). Both outside the VFO box, both wires stay connected to their sense diodes.
- Run with the rocker in **NOR** (both sense diodes reverse biased) + S3 CAL: CLK2 = **2.5 MHz** (D5 load ≈ 40–50 pF, 1 k → RC ≈ 50 ns; 5 MHz would be too close to the T1 limit), counted over 10 PPS s (≈ 0.04 ppm) → CORR saved.
- **Plug the GPSDO 1PPS into J4 pin 9 only for calibration** — the PPS line idles LOW and the LCD would show PRE.

10. Data audio + PTT relay proposal



- Out from PC-300 pin 22: after the AGC, before AF gain and VariBeam → level independent of the volume knob.
- Pole 2 selects the PC-200 pin 17 (mic input) source: **COM** = PC-200 pin 17, **NC** = mic (from the mic socket), **NO** = PC audio (47 k / 470 Ω = line to mic level). RX / mic PTT: mic only; CAT TX: PC audio only. MIC gain sets both levels. Shields joined past the relay, unshielded ends ≤ 2 cm.
- DPDT relay (e.g. Finder 30.22, 16 mA) + flyback diode: PC PTT keys the radio **and** opens the mic line. Mic PTT unchanged.
- **PTT relay board wires: yellow** = VariBeam D9, **red** = +12 V (through PTT ON/OFF), **grey** = GND, **white** = PTT to the KEY line.

- **Driver:** VariBeam D9 → 4.7 k → BC547 base, 47 k base–GND, emitter GND, collector → coil low end. Coil 12 V DPDT, 930 Ω (≈ 13 mA); 1N4148 across the coil, **band to +12 V**.
- **PTT ON/OFF switch** (SPST) in the coil +12 V feed: hard off for uploads / testing; ON for FT8 / data. TX timeout also in the VariBeam firmware.
- VFO Nano D13 → 10 k → VariBeam D8: D8 HIGH ≥ 300 ms keys, LOW releases at once (D13 bootloader blinks ≈ 100 ms are ignored).
- Transformers 600:600; PC side (pin 6) floating, radio side to the radio audio ground.